Pratyush Jha

Pre-doctoral researcher, Indian Institute of Science, Bangalore
pratyushjha254@gmail.com · pratyushjha@iisc.ac.in
Google Scholar · GitHub · LinkedIn · Portfolio Website

Research interests. Physics × machine learning. I use physical systems whose statistics are known exactly or characterized well (spin models, turbulent flows) as controlled testbeds for generative models: measuring how faithfully a model reproduces physical observables, and separating network (learned-score) error from sampler (discretization) error. Longer term, I want to use statistical physics to understand how these models learn, and to carry the methods to quantum many-body and particle physics.

Education

Indian Institute of Technology Roorkee — Nov 2020 – Jul 2024
B.Tech. in Engineering Physics — CGPA 9.117/10 · Department Rank 2

- **Thesis:** *Deep Learning Simulations for Phase Diagrams of Interacting Josephson Qubits* (Advisor: Prof. Peayush Choubey).
- **Physics coursework:** Mechanics & Relativity, Electromagnetic Theory, Thermal & Statistical Physics, Quantum Physics, Quantum Computing, Mathematical Physics, Elements of Condensed Matter Physics, Quantum Theory of Solids†, Physics of Nanosystems, Nuclear Physics & Applications, Nuclear Instrumentation†, Nuclear Astrophysics†.
- **Mathematics, statistics & ML coursework:** Statistical Inference†, Stochastic Processes†, Optimization Techniques, Numerical Analysis & Computational Physics, Signals & Systems, Machine Learning, Reinforcement Learning†, Data Mining for Business Intelligence.
- **Electronics coursework:** Analog Electronics, Digital Electronics & Circuits, Semiconductor Devices, Electrical & Electronic Materials, Applied Instrumentation, Microprocessors & Peripheral Devices.

†Postgraduate-level course.

Publications & Preprints

[1] **P. Jha***, B. Maji*, R. Pandit. *How well can Diffusion Models learn Lagrangian-Tracer Statistics in Non-reciprocal Turbulence?* Preprint, arXiv:2608.27378 (Aug 2026). **Equal contribution.*

[2] **P. Jha**. *Capturing the Topological Phase Transition and Thermodynamics of the 2D XY Model via Manifold-Aware Score-Based Generative Modeling.* Preprint, arXiv:2602.07548 (Feb 2026). *Sole author.*

[3] **P. Jha**. *Coherence at the Financial Misinformation Detection Challenge Task: A Courtroom-Inspired Framework for Reference-Free Verification.* Workshop on Misinformation Detection, ICWSM 2026 (May 2026). *Sole author; 3rd place in the shared task.* DOI: 10.36190/2026.40.

Research Experience

Indian Institute of Science, Bangalore — May 2026 – Present
Project Associate (Pre-doctoral Research) — Advisor: Prof. Rahul Pandit

Generative diffusion models for non-reciprocal turbulence [arXiv:2608.27378]

- To our knowledge, the first application of generative diffusion models to non-reciprocal binary-fluid turbulence (2D Non-Reciprocal Cahn–Hilliard–Navier–Stokes), benchmarking synthetic Lagrangian-tracer trajectories against direct numerical simulations.
- Validated generated trajectories on high-order statistics that probe the distribution tails: structure functions and flatnesses of velocity increments, multiscaling exponent ratios, velocity and acceleration PDFs, curvature statistics, irreversibility and model transferability.
- Sampler dependence: the accelerated DDIM sampler loses precision below 100 steps on some statistics, although it stays accurate down to 25 steps in 3D Navier–Stokes turbulence (Li *et al.*, 2026); an 800-step DDPM reconstructs the flow statistics accurately.
- Argued that anisotropic, intermittent physical flows expose limitations of generative models that standard image and text benchmarks do not reveal.

Follow-up to the 2D XY study [ongoing]

- Roomany–Wyld-style finite-size scaling of correlation lengths on samples from the XY diffusion model.
- Testing Metropolis-adjusted Langevin sampling (MALA) driven by the learned score as an accuracy check: whether converged MALA samples reveal what the score network has actually learned.

Independent Research Jan 2026 – Apr 2026
Sole-author projects

Manifold-aware score-based generative modeling of the 2D XY model [arXiv:2602.07548]

- Separated the two error sources of a diffusion model: the learned score, checked directly against the exact Boltzmann score, and the sampler. The remaining bias in the heat capacity C_v came from ODE discretization, not the network; a MALA (Metropolis-adjusted Langevin) correction after ODE sampling removes it.
- To my knowledge, the first data-driven generative model (no Hamiltonian, no feature engineering) to reproduce the heat capacity of the 2D XY model across the BKT transition.
- Performed score-based diffusion directly on the torus (the 64×64 XY lattice, a 4096-dimensional torus); the learned score matches the Boltzmann score more precisely than a Euclidean-embedding diffusion model, which spends capacity learning the manifold constraint.
- Generalized zero-shot from $L = 8$ to $L = 64$ without retraining, capturing the BKT transition at unseen lattice sizes.

Courtroom-inspired verification of financial misinformation [ICWSM 2026 workshop, DOI: 10.36190/2026.40]

- An LLM (DeepSeek-chat) generates prosecution (falsifying) and defense (justifying) arguments for each claim; a classifier with a frozen ModernBERT encoder, parallel BiGRU layers and attention pooling weighs them, with no external reference documents.
- 3rd place in the FMD Challenge 2026: accuracy 0.955 and F1 0.955 on the blind test set, 38.5% above the baseline accuracy.

Indian Institute of Technology Roorkee Aug 2023 – Apr 2024
Bachelor's Thesis Researcher Advisor: Prof. Peayush Choubey

- Studied how frustrated Josephson-junction arrays map onto Ising-type models with long-range interactions, which motivated using Ising models as a tractable first step.
- Transfer learning toward frustrated Josephson-junction arrays (long-range Ising models, costly for Monte Carlo): pre-trained deep networks on nearest-neighbour (NN) Ising data and fine-tuned them on limited next-nearest-neighbour (NNN) data to predict phase transitions, extending earlier zero-shot NN→NNN transfer.
- Compared fine-tuning configurations across NN and NNN Ising models for predicting phase transitions from limited data.
- Earlier lab project (Jan–Apr 2023): sampled 2D Ising configurations by Monte Carlo and classified their phases with feed-forward and convolutional networks across lattice sizes.

McGill University, Montreal May 2023 – Aug 2023
MITACS Globalink Research Intern Advisor: Prof. John Kildea

- Nuclear-instrumentation project on neutron detection with CR-39 solid-state track-etch detectors, modelled with Monte Carlo particle-transport simulations in TOPAS and TOPAS-nBio (both built on Geant4).
- Simulated recoil protons produced by neutrons and transported through aluminium degraders; reproduced published recoil-proton threshold energies across nine degrader thicknesses.
- Compared secondary-particle dose deposition from an isotropic monoenergetic neutron source with and without a water phantom.

Industry Experience

OLA Krutrim, Bangalore Jul 2024 – Dec 2025
Software Development Engineer I (AI Engineering & Research)

AI engineering

- Built CCTV video-analytics pipelines with the NVIDIA DeepStream SDK over RTSP streams from 1000 service centres.
- Designed a retrieval-augmented generation (RAG) platform in Flask: document ingestion, chunking and Milvus-based similarity retrieval, with Celery and Redis orchestrating parallel tasks.
- Evaluated and deployed embedding and reranking models with KServe and a FastAPI backend at sub-200 ms latency.

AI research

- Fine-tuned LLaMA-3-8B-Instruct with LoRA on 1000+ question–answer pairs built from code, for repository understanding.
- Reduced hallucination after fine-tuning with SimPO preference optimization: a 25% accuracy gain on training questions and 15% on paraphrased questions.
- Investigated reasoning over information internalized from multiple files, through targeted prompting, a literature review and RL-based methods.

Selected Projects

- **Feature-based dynamic pricing with contextual bandits** Feb 2024 – Mar 2024
  Reinforcement Learning course project, Prof. Manu Kumar Gupta
  Implemented the ellipsoid and EXP4 algorithms of Cohen *et al.* (2020) for contextual-bandit pricing and compared their regret, and how they adapt, under changing distributions.

- **Monte Carlo simulation of magnetic skyrmions** Aug 2023 – Oct 2023
  Physics and Astronomy Club, IIT Roorkee
  Metropolis sampling with simulated annealing to form topologically protected skyrmion configurations on a 2D square lattice.
- **MultiPDF Insight Engine: retrieval-augmented generation with Gemini** Aug 2023 – Oct 2023
  Independent project
  Streamlit app answering questions over multiple uploaded PDFs; agentic chunking into propositions with Google Gemini, and Hypothetical Document Embeddings (HyDE) to improve retrieval.
- **Image colorization with conditional GANs** Mar 2023 – Apr 2023
  Machine Learning course project, Prof. Balasubramanian Raman
  Trained a conditional GAN for image-to-image translation in the Lab colour space on 8000+ COCO images.
- **Quantum Hadamard edge detection** Sep 2022 – Nov 2022
  Quantum image processing project, Prof. Peayush Choubey
  Implemented quantum Hadamard edge detection (QHED; Yao *et al.*, 2017): images amplitude-encoded on qubits, with a Hadamard gate on an auxiliary qubit computing differences between neighbouring pixels to extract horizontal and vertical edges.
- **Super Mario reinforcement-learning agent** Jun 2022 – Jul 2022
  Vision and Language Group, IIT Roorkee
  Deep Q-learning agent with a convolutional network in PyTorch that clears the first level of Super Mario Bros.

Skills

Generative modeling	Diffusion models (DDPM, DDIM, probability-flow ODE, score-based models on manifolds); conditional GANs
Language & sequence models	LLM fine-tuning (LoRA), preference optimization (SimPO), transformer encoders (ModernBERT), recurrent networks (BiGRU), retrieval-augmented generation
Reinforcement learning	Deep Q-learning; contextual bandits (ellipsoid method, EXP4)
Scientific computing	Monte Carlo methods (Metropolis, simulated annealing), Langevin dynamics and MALA, ODE samplers; Geant4-based particle transport (TOPAS, TOPAS-nBio); quantum circuits (QHED)
Software	Python, C++, PyTorch, TensorFlow; production ML serving (FastAPI, KServe, NVIDIA DeepStream); Flask, Celery, Redis, Milvus, Streamlit

Selected Awards

- GATE Data Science & AI: All India Rank 117 of 69,242 candidates 2026
- 3rd place (as independent researcher), Reference-Free Financial Misinformation Detection Challenge 2026
- Keynote Speaker, NextGen Indian Leaders Program (DWDG): addressed 200+ attendees on GenAI applications 2024
- MITACS Globalink Research Internship, fully funded research at McGill University 2023
- ENCORE Award for All-Round Excellence, IIT Roorkee Alumni Awards 2022

Teaching & Service

- Student Mentor for 10+ first-year students, IIT Roorkee 2022–2024
- Undergraduate Teaching Assistant for Mathematics-I (70+ first-year students), IIT Roorkee 2021
- Deputy Cell Secretary, NSS IIT Roorkee: led 25 executives and 50 volunteers supporting 100+ underprivileged students; received the 'Best Cell Member' award 2021–2022
- Mentor to a team of 5 students, Business Baazigar case-study competition, ShARE IIT Roorkee 2022

Other Activities

- Clean-energy initiative (CEEW and NSS IIT Roorkee): analysed awareness of energy-efficiency star ratings among Indian consumers and wrote a report for the Bureau of Energy Efficiency.
- Enactus IIT Roorkee: contributed to Project Anushruti, a sign-language translator for students of the Anushruti Academy for the Deaf.
- ShARE IIT Roorkee: leadership and consulting training by former McKinsey and BCG consultants; consulting projects with the startups Curelink and Ridobiko.
- Chess: bronze (team) in the Inter-Bhawan Sports General Championship, IIT Roorkee (3rd of 15 hostels).